

Dokumentasi Software Architecture Lab Project - WSpeedRun.com

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1. Role Pengguna

- Guest
 - Melihat User Profile berdasarkan UserID
 - Melihat daftar keseluruhan game
 - Melihat detail dari sebuah game dengan kategorinya
 - Melihat detail dari sebuah kategori berdasarkan RunCategoryID
 - Melihat leaderboard
 - Hanya dapat melihat detail run dengan status ACCEPTED
 - Melihat detail dari sebuah speedrun berdasarkan RunID
- User
 - Melakukan register dan login
 - Mengirim submit speedrun
 - Melihat keseluruhan riwayat run milik sendiri
 - Mengirim dan menghapus komentar di sebuah run
- Admin
 - Membuat game baru, mengubah detail game yang sudah ada, serta menghapus sebuah game
 - Membuat kategori run, mengubah detail kategori yang sudah ada, serta menghapus sebuah kategori
 - Melihat daftar speedrun berdasarkan status
 - Menerima atau menolak speedrun berdasarkan RunID

2. System Architecture

Sistem dibagi menjadi tiga service utama yang saling berkomunikasi menggunakan HTTP request.

Service Name	Port	Description
Auth Service	3000	Handles user registration, login, authentication (JWT generation), and profile-related operations.
Game Service	3001	Manages game and run category data, including creating, updating, deleting, and retrieving game information.
Run Service	3002	Handles speedrun submissions, verification processes (accept or reject), leaderboard data, and run-related comments.

3. Environment Variables

PORT=3000

DATABASE_URL="mysql://root:@localhost:3306/wspeedrun_db"

JWT_SECRET="Super_Secret_SoftArch"

AUTH_SERVICE_URL=http://localhost:3000
GAME_SERVICE_URL=http://localhost:3001
RUN_SERVICE_URL=http://localhost:3002

4. How to Run the Project

1. Open Project

Open the project folder using Visual Studio Code.

2. Install Dependencies

Each service must install dependencies separately.

```
npm install
```

3. Generate Prisma

Each service must generate prisma separately.

```
npx generate prisma
```

4. Run the Services

Each service must run in separate terminals.

```
npm run start:dev
```

5. Authentication dan Authorization

Authentication menggunakan JWT (JSON Web Token). User akan mendapatkan token setelah berhasil login.

- AuthGuard digunakan untuk endpoint user
- AdminGuard digunakan untuk endpoint admin
- JWT disimpan menggunakan environment variable

6. Endpoint

Method Endpoint	Role	Deskripsi
1. POST/auth/register	Guest	Register akun
2. POST/auth/login	Guest	Login akun untuk mendapatkan JWT Token
3. GET/users/:id/profile	Guest	Melihat User Profile berdasarkan UserID
4. GET/games	Guest	Melihat daftar keseluruhan game
5. GET/games/:id	Guest	Melihat detail dari sebuah game dengan kategorinya berdasarkan GameID
6. POST/admin/games	Admin	Membuat game baru
7. PATCH/admin/games/:id/update	Admin	Mengubah detail game yang sudah ada berdasarkan GameID
8. DELETE/admin/games/:id/delete	Admin	Menghapus sebuah game berdasarkan GameID
9. GET/categories/:id	Guest	Melihat detail dari sebuah kategori berdasarkan RunCategoryID
10. POST/admin/categories	Admin	Membuat kategori run

11. PATCH/admin/categories/:id/update	Admin	Mengubah detail kategori yang sudah ada berdasarkan RunCategoryID
12. DELETE/admin/categories/:id/delete	Admin	Menghapus sebuah kategori berdasarkan RunCategoryID
13. GET/runs/:id/category	Guest	Melihat leaderboard
14. GET/runs/:id/user	Guest, User	Melihat speedrun yang pernah dilakukan User berdasarkan UserID.
15. GET/runs/:id	Guest	Melihat detail dari sebuah run berdasarkan RunID
16. POST/runs	User	Membuat speedrun baru
17. GET/admin/runs/:status	Admin	Melihat daftar speedrun berdasarkan status
18. POST/admin/runs/:id/accept	Admin	Menerima speedrun berdasarkan RunID
19. POST/admin/runs/:id/reject	Admin	Menolak speedrun berdasarkan RunID
20. POST/comments	User	Membuat komentar di sebuah speedrun
21. DELETE/comments/:id	User	Menghapus komentar di sebuah speedrun berdasarkan CommentID

AUTH-SERVICE

1 POST/auth/register

```

Curl
curl -X 'POST' \
  'http://localhost:3000/auth/register' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "JohnDoe",
    "email": "john@mail.com",
    "country": "Indonesia",
    "password": "qWERTy123!"
  }'

```

Request URL

http://localhost:3000/auth/register

Server response

Code	Details
201	Response body <pre> { "message": "Registration successful!", "data": { "username": "JohnDoe", "email": "john@mail.com", "country": "Indonesia", "role": "user" } } </pre>

```

Curl
curl -X 'POST' \
  'http://localhost:3000/auth/register' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Admin",
    "email": "Admin@mail.com",
    "country": "Indonesia",
    "password": "@!@cd121"
  }'

```

Request URL

http://localhost:3000/auth/register

Server response

Code	Details
201	Response body <pre> { "message": "Registration successful!", "data": { "username": "Admin", "email": "Admin@mail.com", "country": "Indonesia", "role": "ADMIN" } } </pre>

2 POST/auth/login

[illegible]

3 GET/users/:id/profile

```
Curl -X GET \
  "http://localhost:3000/users/808bcf8b-958c-497e-8bd7-0f1a3b74007a/profile" \
  -H "accept: /*"
```

Request URL

http://localhost:3000/users/808bcf8b-958c-497e-8bd7-0f1a3b74007a/profile

Server response

Code	Details
200	Response body <pre>{ "username": "michelle", "email": "michelle@gmail.com", "country": "Indonesia", "role": "User" }</pre> Response headers <pre>access-control-allow-origin: * connection: keep-alive content-length: 87 content-type: application/json; charset=utf-8 date: Sun, 24 May 2020 17:58:28 GMT etag: W/"57-15yved7c5k0mVlAe07y0R" keep-alive: timeout=5 powered-by: Express</pre>

GAME-SERVICE

- 4 GET/games
As a guest,

```
Curl
curl -X 'GET' \
  'http://localhost:3001/games' \
  -H 'accept: */*'

Request URL
http://localhost:3001/games

Server response
Code    Details
200

Response body
{
  "game_id": "31a8b4a1-3449-4370-9302-c3b3bde7010",
  "game_name": "Color Sort",
  "description": "Kids-friendly mobile game"
},
{
  "game_id": "12a0f2ed-5770-426f-af04-c2ce142b9914",
  "game_name": "Dreadopia 2.0",
  "description": "A sandbox game focused on building, exploration, and survival."
},
{
  "game_id": "84f80318-91f5-42ff-b262-426acca33cfc",
  "game_name": "Minecraft",
  "description": "A sandbox game focused on building, exploration, and survival."
},
{
  "game_id": "c1e32c11-2a9b-4251-924d-183b28b17c7",
  "game_name": "Candy Crush",
  "description": "Your favorite mobile game"
}
```

- 5 GET/games/:id
As a guest,

```
Curl
curl -X 'GET' \
  'http://localhost:3001/games/c1e32c11-2a9b-4251-924d-183b28b17c7' \
  -H 'accept: */*'

Request URL
http://localhost:3001/games/c1e32c11-2a9b-4251-924d-183b28b17c7

Server response
Code    Details
200

Response body
{
  "game_id": "c1e32c11-2a9b-4251-924d-183b28b17c7",
  "game_name": "Candy Crush",
  "description": "Your favorite mobile game",
  "categories": []
}
```

```
Curl
curl -X 'GET' \
  'http://localhost:3001/games/84f80318-91f5-42ff-b262-426acca33cfc' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJ0dGI0I3UzI3M1c1N0IjC1N0Ijg0PC9yZy9pZC1M1jRkZT'

Request URL
http://localhost:3001/games/84f80318-91f5-42ff-b262-426acca33cfc

Server response
Code    Details
200

Response body
{
  "game_id": "84f80318-91f5-42ff-b262-426acca33cfc",
  "game_name": "Minecraft",
  "description": "A sandbox game focused on building, exploration, and survival.",
  "categories": [
    {
      "run_category_id": "8610e3d-2bfc-4556-babf-7a0112408c94",
      "game_id": "84f80318-91f5-42ff-b262-426acca33cfc",
      "run_category_name": "Minecraft glitchless"
    }
  ]
}
```

- 6 POST/admin/games
As a guest,

```
Curl
curl -X 'POST' \
  'http://localhost:3001/admin/games' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "game_name": "Pizza Maker",
    "description": "Try to make a perfect pizza."
  }'

Request URL
http://localhost:3001/admin/games

Server response
Code    Details
401

Error: Unauthorized
Response body
{
  "message": "Unauthorized. Administrative token is missing.",
  "error": "Unauthorized",
  "statusCode": 401
}
```

As an user,

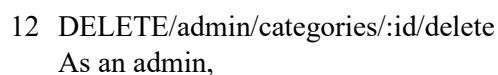
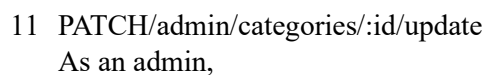
As an admin,

7 PATCH/admin/games/:id/update
As an admin,

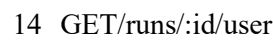
8 DELETE/admin/games/:id/delete
As an admin,

9 GET/categories/:id
As a guest,

10 POST/admin/categories
As an admin,



As a guest,



```
Curl
curl -X "GET" \
  "http://localhost:3002/runs/8d88cf6b-958c-497e-8bd7-0f1a3674007a/user" \
  -H "accept: */*"

Request URL
http://localhost:3002/runs/8d88cf6b-958c-497e-8bd7-0f1a3674007a/user

Server response

Code    Details
200
Response body
{
  "run_id": "120d4ef9-0c53-4f02-81a2-03cfcebf0ba1",
  "run_category_id": "4ba371d1-0a6b-46d4-952b-b372ef7ae293",
  "user_id": "8d88cf6b-958c-497e-8bd7-0f1a3674007a",
  "vod_url": "https://www.youtube.com/watch",
  "submitted_at": "2026-05-23T10:25:39.000Z",
  "verified_at": "2026-05-23T16:17:12.000Z",
  "status": "ACCEPTED",
  "run_duration": "0 Hour(s) 20 Minute(s) 0 Second(s)"
},
{
  "run_id": "120d4ef9-363b-490e-a3f2-6c8dbaf5d55",
  "run_category_id": "8416661d-7c3c-455b-babf-7481248dc94",
  "user_id": "8d88cf6b-958c-497e-8bd7-0f1a3674007a",
  "vod_url": "https://www.youtube.com/watch?v=0p6m2mgXUQ",
  "submitted_at": "2026-05-23T10:16:31.000Z",
  "verified_at": "2026-05-23T16:16:53.000Z",
  "status": "ACCEPTED",
  "run_duration": "1 Hour(s) 0 Minute(s) 0 Second(s)"
}
```

[illegible]

As a guest,

```
curl -X 'GET' \
  'http://localhost:3002/rans/cbb510e-3011-4ccc-903f-e8f10a5dfde' \
  -H 'accept: */*'

Request URL:
http://localhost:3002/rans/cbb510e-3011-4ccc-903f-e8f10a5dfde

Server response

Code      Details
200

Response body
{
  "run_id": "cbb510e-3011-4ccc-903f-e8f10a5dfde",
  "vid_url": "https://www.youtube.com/watch",
  "run_duration": "0 hour(s) 30 Minute(s) 0 Second(s)",
  "status": "PENDING",
  "submitted_at": "2020-05-23T10:10:21.000Z",
  "verified_at": null,
  "run_category_name": "75K",
  "game_information": {
    "game_id": "11ad6d61-3449-4520-9302-c1b10de7632d",
    "game_name": "Galaxy Surf",
    "description": "Kids-friendly mobile game"
  },
  "comments": [],
  "runner_information": {
    "username": "michelle",
    "email": "michelle@mail.com",
    "country": "Indonesia",
    "role": "USER"
  }
}
```

As a guest,

```
curl -X 'POST' \
  'http://localhost:3002/runs' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "run_category_id": "61d2f21b-6366-427a-b5a5-e12734bda981",
    "vod_url": "http://www.youtube.com/watch?v=...",
    "run_duration": 1264
  }'
```

Request URL

http://localhost:3002/runs

Server response

Code	Details
401	Error: Unauthorized

Response body

```
{
  "message": "Unauthorized. Please provide a valid token.",
  "error": "Unauthorized",
  "statusCode": 401
}
```

As an user,

Curl

```
curl -X 'GET' \
  "http://localhost:3002/admin/rules/PENDING" \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ1IiwiaWF0IjoiMTY1MjE1MjE1In0='
```

Request URL

http://localhost:3002/admin/rules/PENDING

Server response

Code Details

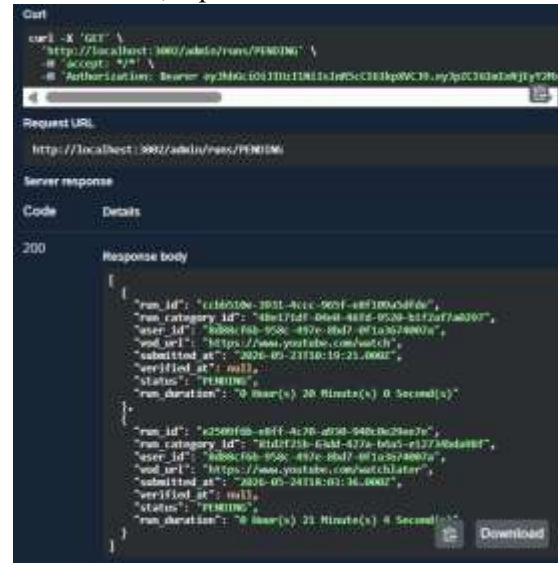
403 Error: Forbidden

Response body

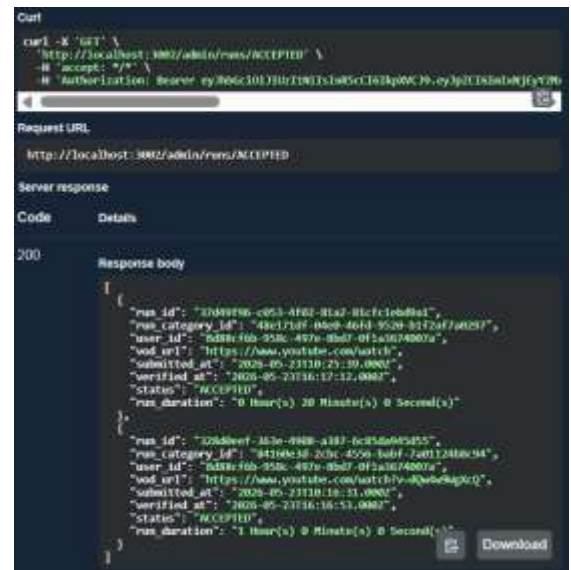
```
{
  "message": "Forbidden. Only users with an ADMIN role can access this endpoint.",
  "error": "Forbidden",
  "statusCode": 403
}
```

Download

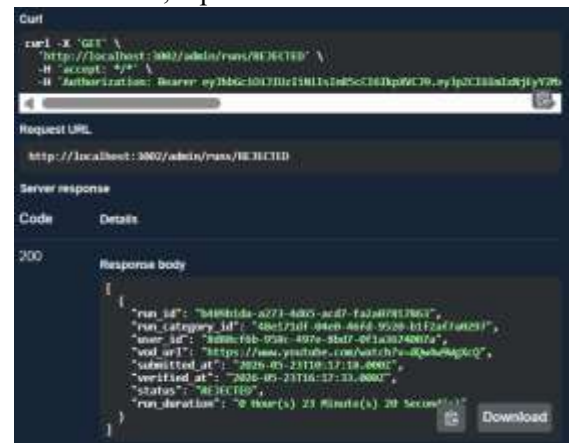
As an admin, input : PENDING



As an admin, input : ACCEPTED

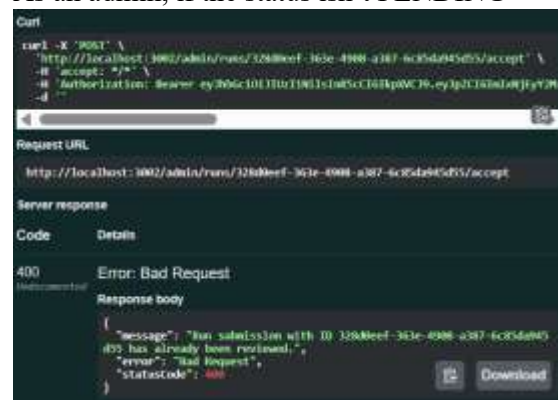


As an admin, input : REJECTED



18 POST/admin/runs/:id/accept

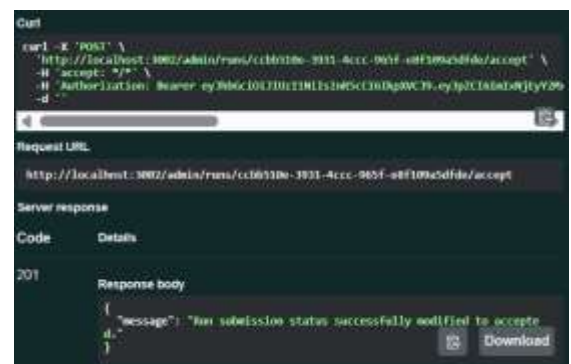
As an admin, if the status isn't PENDING

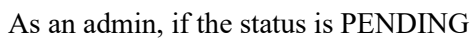


As an admin, if the status is PENDING

19 POST/admin/runs/:id/reject

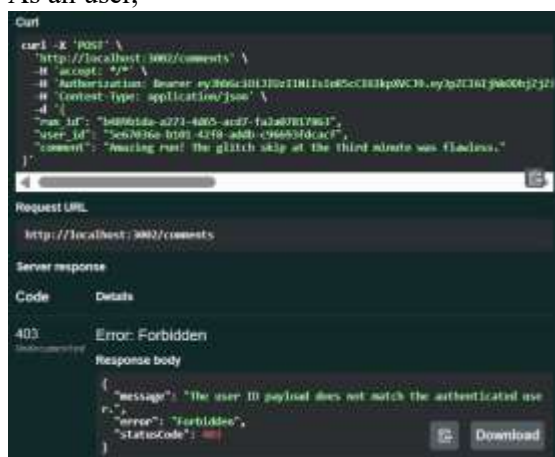
As an admin, if the status isn't PENDING





21 DELETE/comments/:id

As an user,



As an authenticated user,

[illegible]